

GVPT Data Science Camp

Data Visualization

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University of Maryland

- ▶ Introduction to using `R` and `RStudio`

Learning Objectives

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- ▶ Create your first plot in `R`

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- ▶ Introduction to using **R** and **RStudio**
- ▶ Create your first plot in **R**
- ▶ Test your hypotheses using data visualizations

- ▶ R evaluates the code you type — try some arithmetic:

```
1 + 2
```

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- ▶ Functions take arguments inside parentheses:

```
sum(1, 2)
```

Exercise

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- ▶ What is 48 divided by 5?

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- ▶ What is 48 divided by 5?
- ▶ What is the square root of 5232? HINT: use the `sqrt()` function

- Two ways to get the sum:

```
sum(412, 65, 292)
```

```
#> [1] 769
```

```
412 + 65 + 292
```

```
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```

- ▶ Two ways to get the sum:

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- ▶ Division:

```
48 / 5  
#> [1] 9.6
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sum(412, 65, 292)
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412 + 65 + 292
#> [1] 769
```

- ▶ Division:

```
48 / 5
#> [1] 9.6
```

- ▶ Square root:

```
sqrt(5232)
#> [1] 72.33257
```

- ▶ R has a large number of built-in functions that allow you to manipulate data

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- ▶ For example, the `seq()` function provides all numbers in a specified sequence:

```
seq(from = 1, to = 20, by = 3)  
#> [1]  1  4  7 10 13 16 19
```

- ▶ R has a large number of built-in functions that allow you to manipulate data
- ▶ For example, the `seq()` function provides all numbers in a specified sequence:

```
seq(from = 1, to = 20, by = 3)
#> [1]  1  4  7 10 13 16 19
```

- ▶ Arguments are positional, so naming them is optional:

```
seq(1, 20, 3)
#> [1]  1  4  7 10 13 16 19
```

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- ▶ Install the relevant package(s) once:

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install.packages("tidyverse")
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- ▶ Load the packages in your current session:

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library(tidyverse)
```

- ▶ Using the RStudio console, install the `tidyverse` packages

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- ▶ Using the RStudio console, install the **tidyverse** packages

```
install.packages("tidyverse")
```

- ▶ Load these packages in your current session, along with **readxl** (installed with the tidyverse) for reading Excel files

```
library(tidyverse)  
library(readxl)
```

- ▶ *For you and your co-author's sake*

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- ▶ Keep everything:

Workflow When Using RStudio

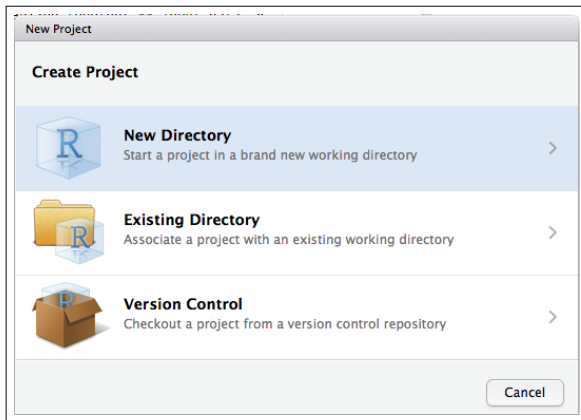
- ▶ *For you and your co-author's sake*
- ▶ Keep everything:
- ▶ Organized

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- ▶ Keep everything:
- ▶ Organized
- ▶ Reproducible

- ▶ *For you and your co-author's sake*
- ▶ Keep everything:
- ▶ Organized
- ▶ Reproducible
- ▶ Annotated

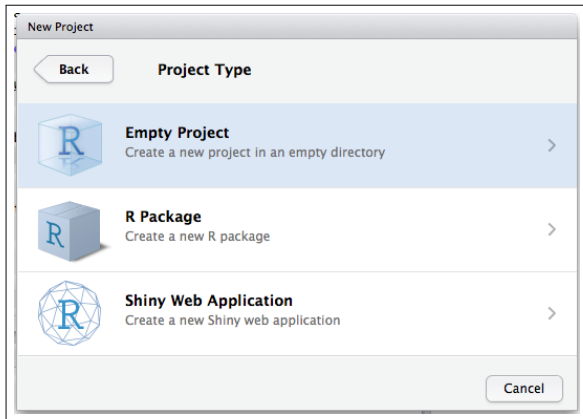
In the RStudio console, type `getwd()` to locate where you are in your computer

Create a new RStudio project for this Data Science Camp



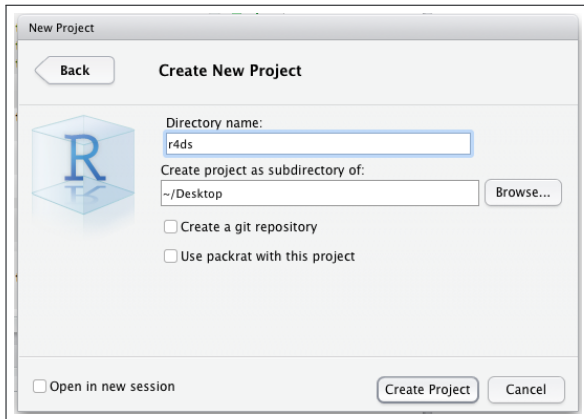
Source: *R for Data Science* (R4DS)

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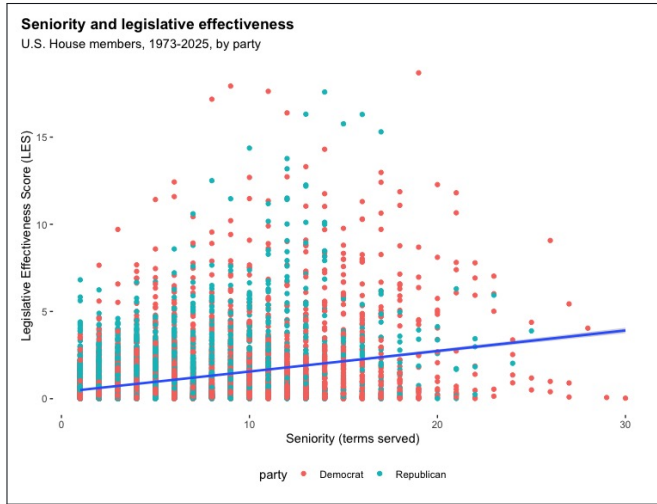
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Re-run `getwd()` to locate where you are now in your computer

- ▶ Data visualization is the graphical representation of information and data

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- ▶ From the Center for Effective Lawmaking (CEL) data: Do more senior members of Congress pass more legislation than newer members?

Skipping to the End



How Did We Do This?

R code:

```
ggplot(data = cel, mapping = aes(x = seniority, y = les)) +  
  geom_point(mapping = aes(color = party)) +  
  geom_smooth(method = "lm") +  
  theme(legend.position = "bottom",  
        panel.grid = element_blank(),  
        panel.background = element_blank(),  
        plot.title.position = "plot",  
        plot.title = element_text(face = "bold")) +  
  labs(title = "Seniority and legislative effectiveness",  
        subtitle = "U.S. House members, 1973-2023, by party",  
        x = "Seniority (terms served)",  
        y = "Legislative Effectiveness Score (LES)")
```

- ▶ Load the relevant packages:

```
library(tidyverse)
library(readxl)
```

Load Relevant Packages and Data

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```
library(tidyverse)
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```

- ▶ Download the Center for Effective Lawmaking House dataset ([CELHouse93to118](#)) from thelawmakers.org

Load Relevant Packages and Data

- ▶ Load the relevant packages:

```
library(tidyverse)
library(readxl)
```

- ▶ Download the Center for Effective Lawmaking House dataset ([CELHouse93to118](#)) from thelawmakers.org
- ▶ Save it in your project folder as [CELHouse93to118.xlsx](#)

- ▶ Read the Excel file into an object named `cel`:

```
cel <- read_excel("CELHouse93to118.xlsx")
```

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```
cel <- read_excel("CELHouse93to118.xlsx")
```

- ▶ One row per member per Congress: every member of the U.S. House from the 93rd Congress (1973) through the 118th (2023)

Rename the Variables

The columns arrive with long descriptive names — give them short ones. Copy this chunk as is; we will unpack how it works tomorrow:

```
cel <- rename(cel,
  member      = "Legislator name, as given in THOMAS",
  congress    = "Congress number",
  year        = "Year at start of Congress",
  state       = "Two-letter state code",
  dem         = "1 = Democrat",
  female      = "1 = female",
  vote_pct    = "Percent vote received to enter this Congress",
  ideology    = "First-dimension DW-NOMINATE score",
  seniority   = "Seniority, number of terms served counting current",
  born        = "Year born",
  les         = "LES 1.0",
  bills_passed = "Total number of standalone bills passing House",
  benchmark   = "1 = Below, 2 = Meets, 3 = Above, Based on Classic LES")
```


Keep only the renamed columns, and add readable labels for party and for the CEL benchmark categories:

```
cel <- select(cel, member, congress, year, state, dem, female,
              vote_pct, ideology, seniority, born, les,
              bills_passed, benchmark)

cel <- mutate(cel,
  party = if_else(dem == 1, "Democrat", "Republican"),
  benchmark = factor(benchmark,
                     labels = c("Below", "Meets", "Exceeds")))
```

- ▶ Learn more about this dataset by browsing the CEL codebook at thelawmakers.org

- ▶ Learn more about this dataset by browsing the CEL codebook at thelawmakers.org
- ▶ Get a quick overview of every variable in your console:

```
glimpse(cel)
```

- ▶ Open the data in a spreadsheet-style viewer:

```
view(cel)
```

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```
view(ce1)
```

- ▶ A couple of useful variables — **seniority**: number of terms served, counting the current one; **les**: Legislative Effectiveness Score, a summary of how successful a member is at moving their bills forward

- How many rows and columns are in `cel`?

```
nrow(cel)  
ncol(cel)
```

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```
nrow(cel)
ncol(cel)
```

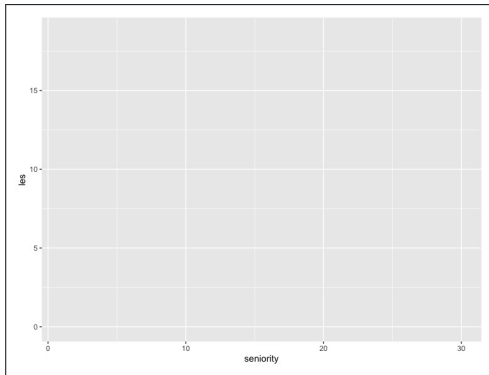
- ▶ What does the `vote_pct` variable describe?

Set Up Your Plot

```
ggplot(data = cel)
```

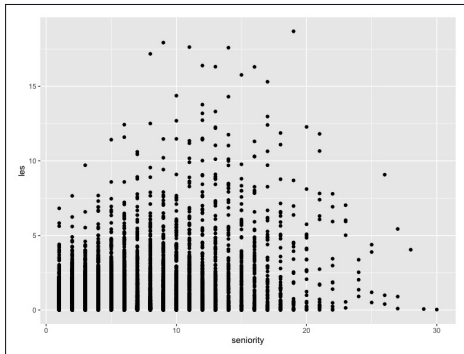



```
ggplot(data = cel, mapping = aes(x = seniority, y = les))
```



Add in Your Legislators

```
ggplot(data = cel, mapping = aes(x = seniority, y = les)) +  
  geom_point()
```



- ▶ Inspect the `les` and `vote_pct` variables and describe them

- ▶ Inspect the `les` and `vote_pct` variables and describe them
- ▶ Make a scatter plot of them

- ▶ Why does the following code give an error, and how would you fix it?

```
ggplot(data = cel) +  
  geom_point()
```

Let's Look at Groups in the Data

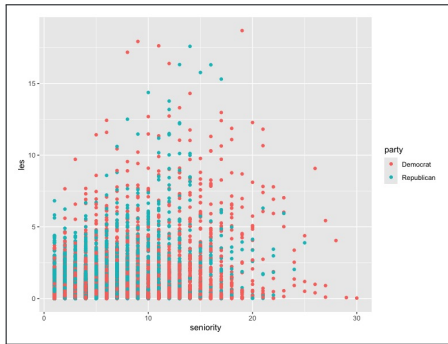
- ▶ You can use visual elements or aesthetics (`aes`) to communicate your data

Let's Look at Groups in the Data

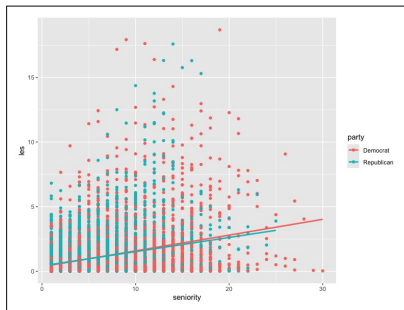
- ▶ You can use visual elements or aesthetics (`aes`) to communicate your data
- ▶ Let's look at a categorical variable: the party of the member (Democrat or Republican)

Let's Plot Groups in the Data

```
ggplot(data = cel,  
       mapping = aes(x = seniority, y = les, color = party)) +  
  geom_point()
```

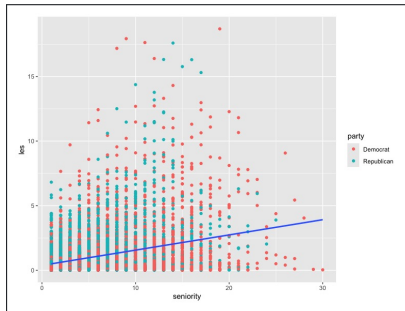



```
ggplot(data = cel,  
       mapping = aes(x = seniority, y = les, color = party)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE)
```



Look at the Relationship Among All Members

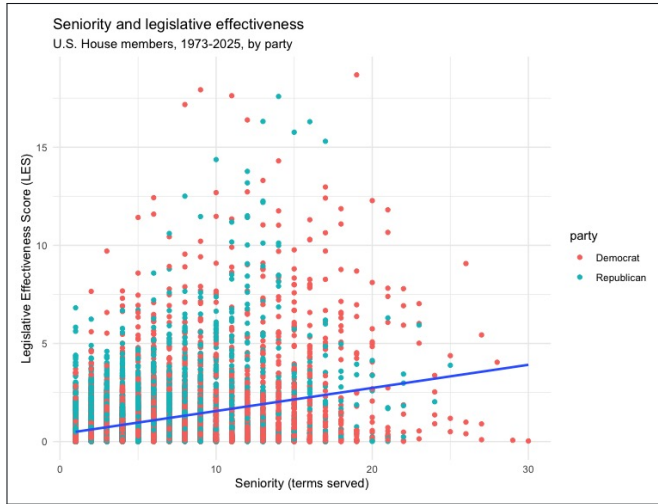
```
ggplot(data = cel, mapping = aes(x = seniority, y = les)) +  
  geom_point(mapping = aes(color = party)) +  
  geom_smooth(method = "lm", se = FALSE)
```



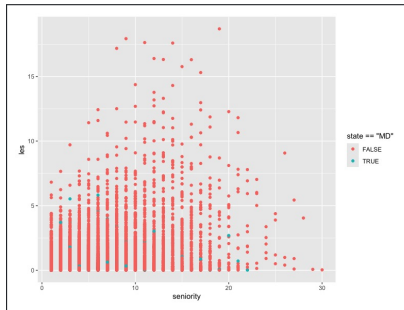
Add Titles and Labels

```
ggplot(data = cel, mapping = aes(x = seniority, y = les)) +  
  geom_point(mapping = aes(color = party)) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(  
    title = "Seniority and legislative effectiveness",  
    subtitle = "U.S. House members, 1973-2023, by party",  
    x = "Seniority (terms served)",  
    y = "Legislative Effectiveness Score (LES)"  
  )
```

Output with Titles and Labels

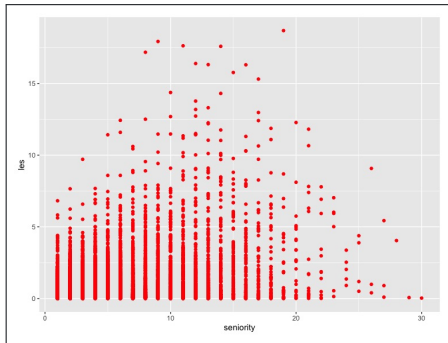


```
ggplot(data = cel) +  
  geom_point(mapping = aes(x = seniority, y = les,  
                           color = state == "MD"))
```



Editing Plots

```
ggplot(data = cel) +  
  geom_point(mapping = aes(x = seniority, y = les), color = "red")
```



- What is wrong with this code?

```
ggplot(data = cel) +  
  geom_point(mapping = aes(x = seniority, y = les, color = "blue")))
```

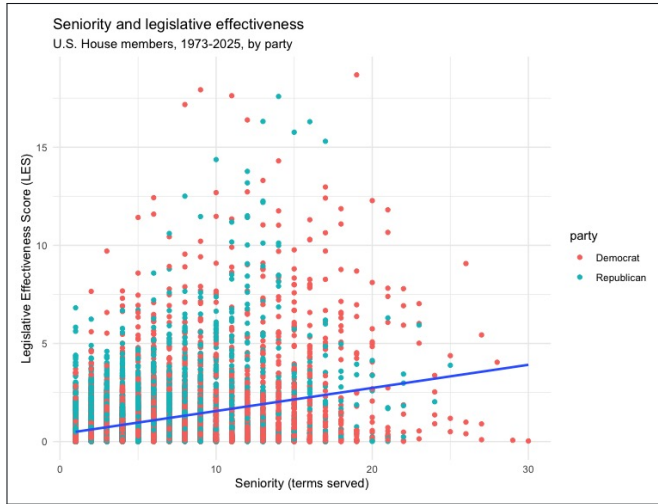
- ▶ Name a categorical and continuous variable in `cel`

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- ▶ Map a continuous variable (try `votepct`) to color. How do the aesthetics behave differently for categorical and continuous variables?
- ▶ Map `state` to the shape aesthetic. What does the warning say?

```
ggplot(data = cel, mapping = aes(x = seniority, y = les)) +  
  geom_point(mapping = aes(color = party)) +  
  geom_smooth(method = "lm", se = FALSE) +  
  theme_minimal() +  
  labs(  
    title = "Seniority and legislative effectiveness",  
    subtitle = "U.S. House members, 1973-2025, by party",  
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    y = "Legislative Effectiveness Score (LES)"  
  )
```

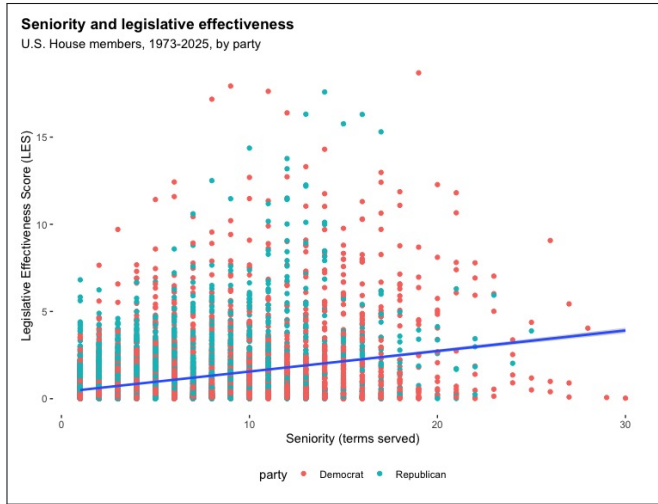
Cleaning our Graph



Creating a Unique Theme

```
ggplot(data = cel, mapping = aes(x = seniority, y = les)) +  
  geom_point(mapping = aes(color = party)) +  
  geom_smooth(method = "lm") +  
  theme(  
    legend.position = "bottom",  
    panel.grid = element_blank(),  
    panel.background = element_blank(),  
    plot.title.position = "plot",  
    plot.title = element_text(face = "bold")  
  ) +  
  labs(  
    title = "Seniority and legislative effectiveness",  
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    x = "Seniority (terms served)",  
    y = "Legislative Effectiveness Score (LES)"  
  )
```

Creating a Unique Theme



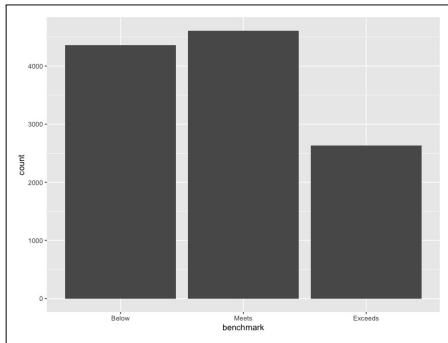
- ▶ Customize the scatter plot you created using the `theme()` argument

- ▶ When visualizing and reporting categorical data, we must use different tools

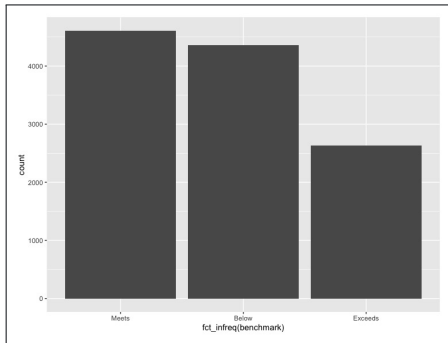
```
select(cel, member, state, party)
```



```
ggplot(cel, aes(x = benchmark)) +  
  geom_bar()
```

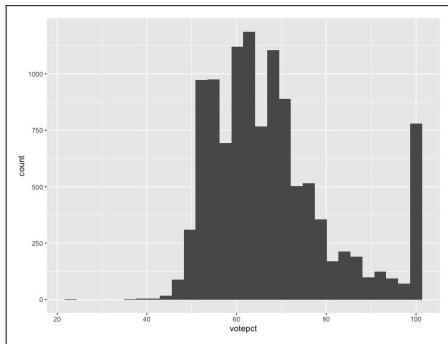


```
ggplot(cel, aes(x = fct_infreq(benchmark))) +  
  geom_bar()
```

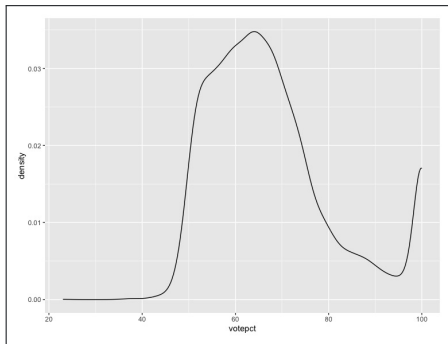


Visualizing Numerical Data

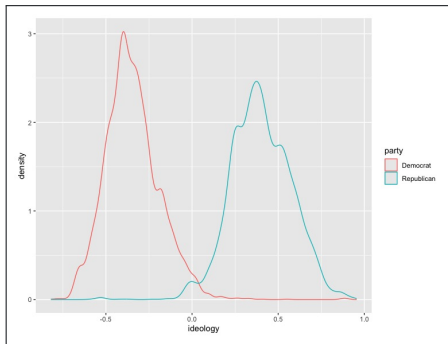
```
ggplot(cel, aes(x = vote_pct)) +  
  geom_histogram()
```



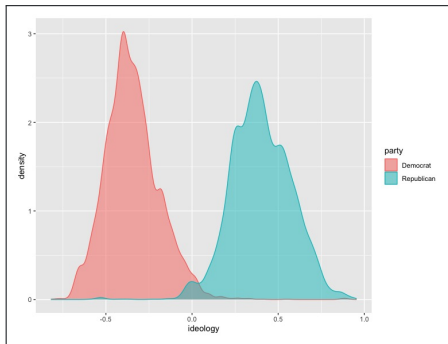
```
ggplot(cel, aes(x = vote_pct)) +  
  geom_density()
```



```
ggplot(cel, aes(x = ideology, color = party)) +  
  geom_density()
```



```
ggplot(cel, aes(x = ideology, color = party, fill = party)) +  
  geom_density(alpha = 0.5)
```



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- ▶ Discovered relationships in the data
- ▶ Connected these relationships to expectations (hypotheses)

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- ▶ Is relevant to your research interests
- ▶ Contains continuous and categorical variables

Questions?